

1 Introduction

1.1 Project Summary

1.1.1 Objectives

[These should describe the overall goal in developing the product, high level descriptions of what the product will do, how they are aligned to business objectives, and the requirements for interaction with other systems.]

To create a new web based Internal Application named sapphire (zenith.sapphire.com) for handling customer queries enabling centralized ticket management system for tracking real time workflow and for auditing SLA compliance

1.1.2 Background

[Provide a brief history of how the project came to be proposed and initiated, including the business issues/problems identified, and expected benefit of implementing the project/developing the product.]

Currently, all the customer queries are handled manually by recording the queries in MS Excel which makes tracking and measuring customer service agents' efficiency difficult.

1.2 Business Drivers

[List the business drivers that make development of this product important. These can be financial, operational, market or environmental.]

- Sapphire enables or handle high volume of customer queries which ultimately reduces response time towards resolving ticket.
- Sapphire act as a centralized tracking system and so none of the customer queries go unresolved.
- y resolving customer queries, it creates trust in the company and attracts more prospects.
- Sapphire also helps to identify the frequently occurring issue so that the management can fix it with the suppliers thereby preventing its further occurrence

1.3 Project Scope

[Describe what work is in scope for the project, and specifically what work is out of scope... beyond the current budget, resources and timeline as approved by the project stakeholders. This is designed to prevent "scope creep" of additional features and functions not originally anticipated.]

1.3.1 In Scope Functionality

- A new web based internal application to handle customer queries that support four roles such as Customer Service Agents, Service Centres, Team Lead and Admin to be launched.
- Call Routing through IVR functionality to be implemented.
- Ticket Creation and SLA tracking to be enabled
- Integration with notification gateways such as message and whatsapp to be enabled

1.3.2 Out of Scope Functionality

- Current process of manual SLA tracking will be out of scope.
- Receiving customer queries through multiple channel except phone calls will be out of scope.
- Notification through emails to all the actors will be out of scope.
- Call recording will be out of scope.

1.4 System Perspective

[Provide a complete description of the factors that could prevent successful implementation or accelerate the projects, particularly factors related to legal and regulatory compliance, existing technical or operational limitations in the environment, and budget/resource constraints.]

1.4.1 Assumptions

- The Sapphire system will reduce customer query response time by **at least 70%** compared to the current manual Excel-based process.
- Customer Service Agents (CSA) and other users will receive **adequate training** to effectively use the Sapphire application.
- The existing **server infrastructure and database** will be capable of supporting the new application without major upgrades.
- The IVR system and messaging gateways will provide **reliable APIs** for seamless integration with the Sapphire platform.

1.4.2 Constraints

- Sapphire must operate on the **existing server and database infrastructure** without requiring major additional hardware investment.
- The system must support **150–200 concurrent users** without performance degradation.
- Integration must be done with the **existing IVR platform**, which may limit flexibility in call routing features.
- The project must be completed within the **approved budget and timeline** set by project stakeholders.

1.4.3 Risks

- Approval delays or additional costs associated with **third-party SMS and WhatsApp messaging gateways**.
- Possible **integration failures** between Sapphire and the IVR platform due to API incompatibility.
- System performance issues if the number of users exceeds the **planned capacity of 200 users**.
- Resistance from customer service agents in **adopting the new system**, which may slow implementation.

1.4.4 Issues

- Customer calls may not reach Customer Service Agents if the **IVR system experiences downtime**.
- Incorrect customer contact information entered by CSAs may lead to **failed notification delivery**.
- High call volumes may temporarily **overload the system**, delaying ticket creation or routing.
- Users may experience **initial difficulty adapting to the new workflow** compared to the manual Excel-based process

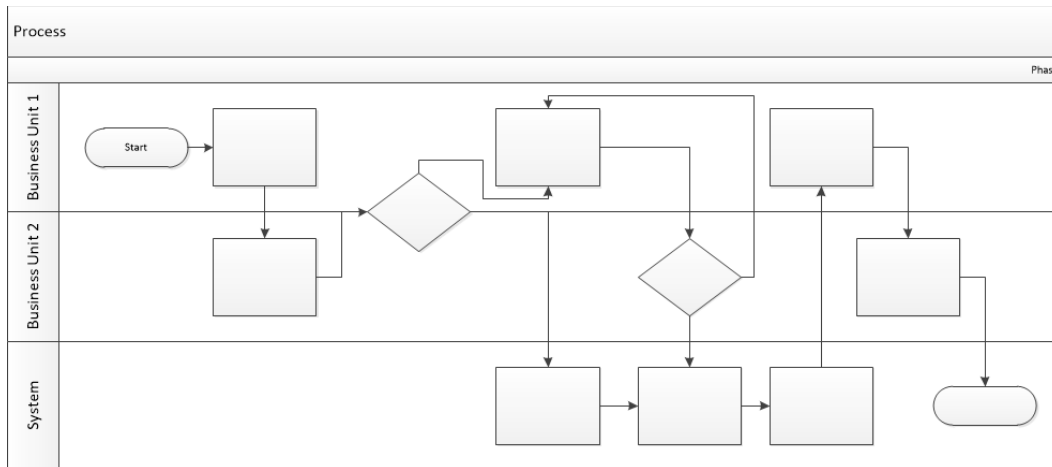
2 Business Process Overview

[Describe how the current process(es) work, including the interactions between systems and various business units. Include visual process flow diagrams to further illustrate the processes the new product will replace or enhance.]

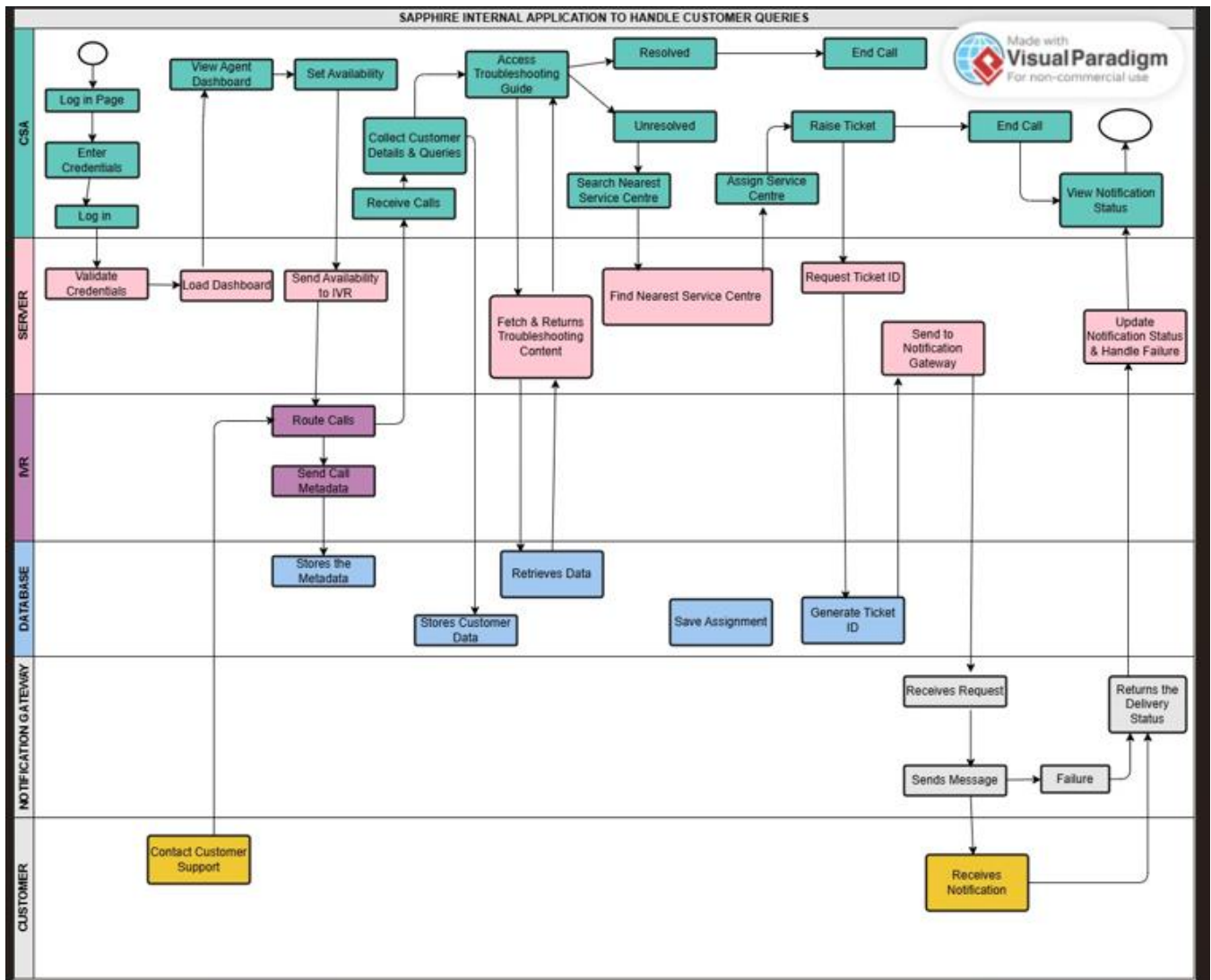
Use case documentation and accompanying activity or process flow diagrams can be used to create the description(s) of the proposed or “To-Be” processes.]

2.1 Current Business Process (As-Is)

There is no current process. This is completely new process.



2.2 Proposed Business Process (To-Be)



3 Business Requirements

[The specific business requirements elicited from stakeholders should be listed, categorized by both priority and area of functionality to smooth the process of reading and tracking them. Include links to use case documentation, and other key reference material as needed to make the requirements as complete and understandable as possible. You may wish to incorporate the functional and non-functional requirements into a traceability matrix that can be followed throughout the project.]

The requirements in this document are prioritized as follows:

Value	Rating	Description
1	High	This requirement is high priority, but the project can be implemented at a bare minimum without this requirement.
2	Medium	This requirement is somewhat important, as it provides some value but the project can proceed without it.
3	Low	This is a low priority requirement, or a “nice to have” feature, if time and cost allow it.

3.1 Business Requirements

BR-001 Admin on board Customer Service Agent:

- 1) Admin will be able to add new customer service agent into the sapphire application.
- 2) Admin will verify the employee and upload the user profile such as employee name, role, team to which the employee belong into the system.
- 3) Only authorized admins can create user profiles in the system.
- 4) Based on the user profile entered, the system will generate credentials such as user name and password which will then be disclosed to the employee.

BR-002 Customer Service Agent Log-in:

- 1) Onboarded Customer Service Agents logs in to the system securely.
- 2) The system provides authority to change availability status. When set to available, CSA will receive calls. If not, CSA will not receive calls.

BR-003 IVR Call Linkage:

- 1) Allows sapphire to integrate with IVR, receive calls from IVR and route call to the available CSA.
- 2) Unique ID will be generated for each call for tracking purpose
- 3) SLA tracker starts to run once the CSA receives the call

BR-004 Ticket Raise:

- 1) CSA will be able to input customer details and queries related to products
- 2) CSAs will also be given access to troubleshooting guides
- 3) If the issue is resolved, CSA will raise a ticket, update the status and end the call

BR-005 Assign Service Centre:

- 1) If the issue remains unresolved even after troubleshooting, the CSA will search the service centre located nearest to the customer location.
- 2) CSA will then assign the ticket to the service centre and terminate the call.
- 3) The respective service centre will be assigned the ticket.

BR-006 Ticket Closure:

- 1) Enables the team lead to approve the closure of ticket after ensuring that the issue is resolved
- 2) Team lead can view the time taken to resolve issues and If SLAs are met

3.2 Functional Requirements

BR-001 Admin on board Customer Service Agent:

FR:01 Admin On-Board Customer Service Agent (CSA):

1. The admin will access the sapphire application by using their login credentials provided to them.
2. Upon successful login, the employee details page must be opened.
3. Admins will manually verify the Customer Service Agent's details.
4. Admin should enter the CSA's Name in "Name Text Box Field", the Role as Customer Service Agent in "Role Text Box Field", the Team Lead of the respective customer service agent in "Team Lead Text Box Field", the shift or working hours of the respective CSA in the "Shift Timing Text Field", the product expertise of the respective CSA from the drop down box, the mobile number of the CSA in "Mobile Number Text Field"
5. In Mobile Number Field, entry should be numeric. If any other character is used, error message should appear to the user.
6. Admin should click the button "Confirm" after all the details are entered.

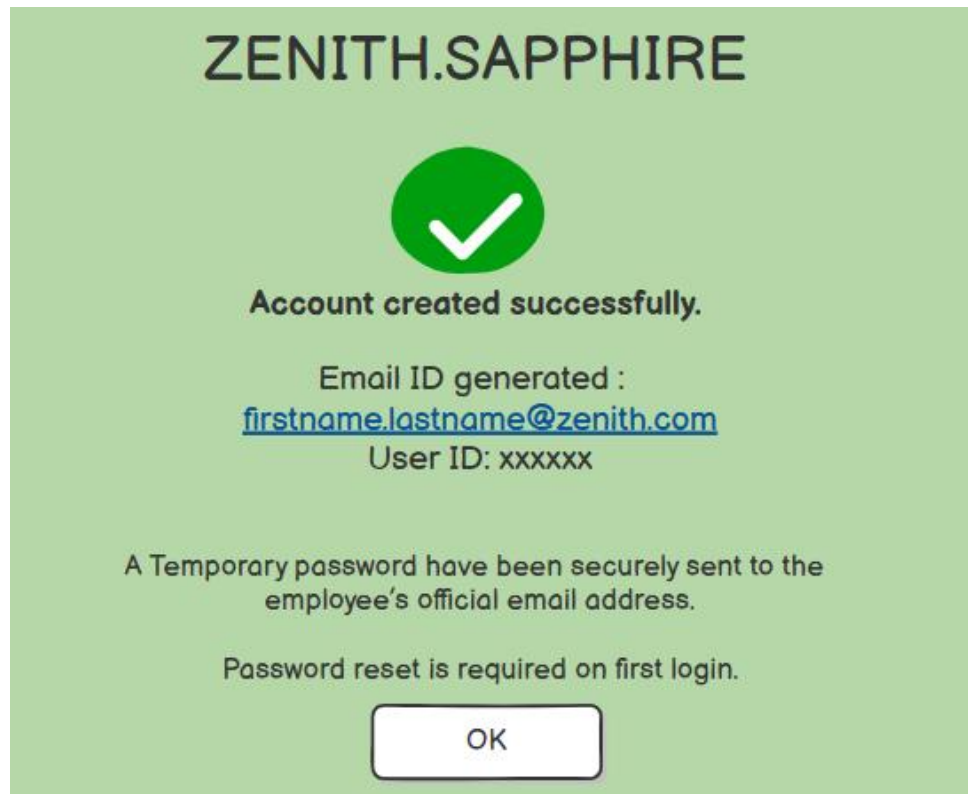
The screenshot shows a web form titled "ZENITH.SAPPHIRE" with a subtitle "Employee Details". The form is set against a light green background. It contains several input fields and a dropdown menu arranged in two columns. The left column includes fields for "Name of the Employee", "Date of Joining", "Date of Birth", and "Role". The right column includes fields for "Mobile Number", "Shift Timings" (a dropdown menu), "Team Leader", and "Product Expertise" (a search-enabled dropdown menu). The "Product Expertise" dropdown is currently open, showing a list of options: "Mobiles", "Laptops", and "Accessories". At the bottom center of the form is a "Confirm" button.

7. Upon clicking the Confirm button, the system must validate all the entered fields.

FR-002 Customer Service Agent Log in & Password Creation:

1. After successful validation, the system must generate an email ID, display it to the admin and store it in the database under customer agent profile
2. The email ID must contain the name of the customer service agent with the company domain. Ex: firstname.lastname@zenith.com
3. In case of similar name, the system must include a numeric identifier to ensure uniqueness. Ex: firstname.lastname21@zenith.com
4. Along with email ID, the system must generate a unique user ID that will be used to log in sapphire application.

5. The user ID must be in alpha numeric format which must contain 6 characters and the same must be stored in database.
6. The format of the temporary password should contain 8 characters of which at least 1 must be upper case letter, 1 must be lower case letter, 1 number and 1 special case.
7. The system should generate a temporary password which must be used to access both official email and sapphire application.
8. The temporary password must be delivered to the provided mobile number of the CSA .
9. The CSA must log in to the official email provided to reset the temporary password for both email and sapphire application
10. The CSA will enter the new password in the "Password Textbox field"
11. The newly entered password must contain 8 characters which must contain at least
 - One uppercase letter
 - One lowercase letter
 - One number and
 - One special character
12. Once the criteria is met, the CSA must enter the same password in the text field "CONFIRM PASSWORD"
13. If the characters of "Password Text Field" and "Confirm password Textbox field" match, the password must be encrypted and stored in database.

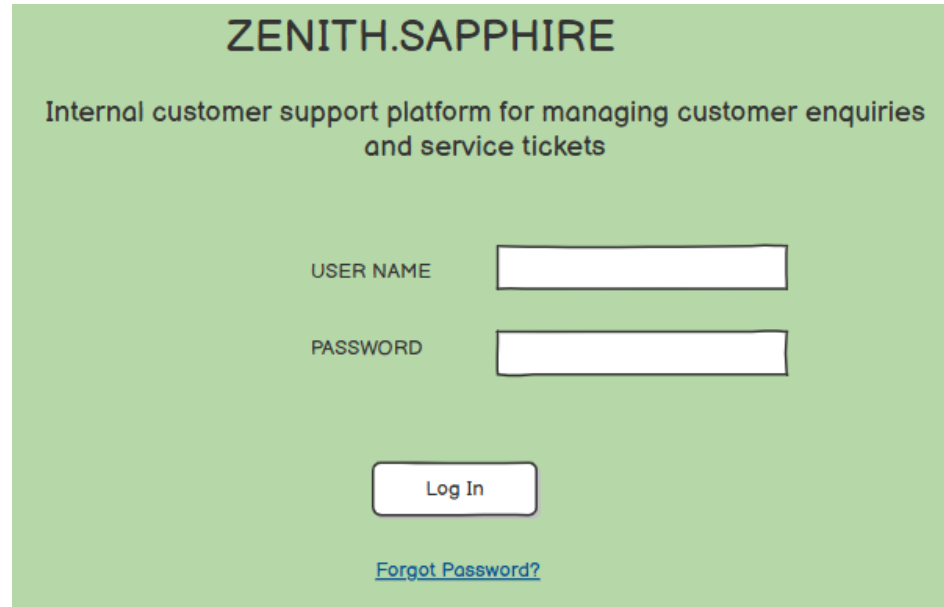


BR-002 Customer Service Agent Log-in

FR-003 Customer Service Agent Login:

1. Customer Service Agent must enter the site zenith.sapphire.com
2. The CSA must see the Log in page.
3. The CSA should enter their user ID in the "Username Text field", the newly generated password in the "Password Text field" and must click log in.
4. The server must verify the credentials entered by the CSA
5. After verification, the server must return the dashboard page to the CSA.
6. If the verification is failed, the system should return an error message and must not provide access to the CSA.

7. In case of failure, retry options must be given five times. And on 5th failed attempt, the user ID must be locked.
8. Apart from retry options, the CSA should also be given an option to reset their password through “Forgot Password” link.
9. Upon clicking “Forgot password” link, the CSA must receive a reset link to their official mail ID.



ZENITH.SAPPHIRE

Internal customer support platform for managing customer enquiries and service tickets

USER NAME

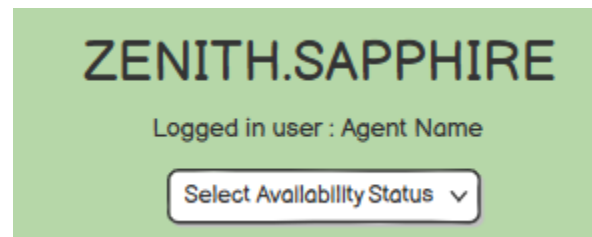
PASSWORD

Log In

[Forgot Password?](#)

FR-004 Customer Service Agent Status Update:

1. When the CSA’s log in attempt is successful, the dashboard page of the respective CSA must appear.
2. The page should also display the CSA name as logged in user.
3. A dropdown box labelled “Availability Status” with two options such as Available and Not available must appear as shown in the picture below



ZENITH.SAPPHIRE

Logged in user : Agent Name

Select Availability Status ▼

4. The CSA must have the option to choose either “Available” or “Not Available”
5. If the option “Available” is set by the CSA, the server must send the status as available to IVR.

BR-003 IVR Call Linkage:

FR-005 Customer Call Lined in Sapphire IVR:

1. When a customer calls, the IVR must route the calls on two conditions. One is to identify the list of CSAs belonging to the product category given by the customer in IVR and the other is to check the availability status of CSA.
2. If both the conditions are met, the IVR will route the call to a CSA based on the following priority rule:
Priority Rule: To the CSA who is available for a longest time
3. If any CSA is available but does not belong to the product category selected by the customer, then the CSA will not be eligible to receive calls.
4. In case of unavailability of any agents , then IVR must place the call in queue and must give busy message to the customer.
5. If the call is routed and the CSA did not the respond, the call must be logged as incomplete.
6. The CSA's dashboard must be notified for an incoming call through a pop up.
7. The CSA must hear the customer on the headset.

FR-006 Unique Call ID generated:

1. The server must generate a unique caller ID which should be in the prescribed format as “yyyymmdd-aabb-ccc-dddd”
 - a) yyyymmdd – Signifies the date (yyyy denotes the year;mm denotes the month; dd denotes the day). Ex:20260506
 - b) aabb – Signifies the time of the day in 24 hour format Ex:14:55
 - c) ccc – Signifies the CSA ID number. Ex: 012
 - d) dddd – Signifies the call sequence for that day Ex: If the call is received as 4th call for the day then it must be numbered as 0004.
2. The server must capture the call details associated with that particular caller ID and store it in the database. The details must include caller ID, call start and call end time, CSA to whom the call was routed.

BR-004 Ticket Raise:

FR-007 Capture Customer Information:

1. Upon receiving the customer call, the CSA must click the customers button from the sidebar and input the following customer details in the respective textbox field provided as shown in the below mentioned picture.

2. The name of the customer to be entered by the CSA in the textbox field labelled as “CUSTOMER NAME”
3. The email ID of the customer to be entered by the CSA in the textbox field labelled as “EMAIL”
4. Mobile Number of the customer to be entered by the CSA in the textbox field labelled as “Mobile Number”
5. The product type or category for which the customer raises the issue must be entered in the textbox field labelled as “PRODUCT TYPE”
6. Brand of the product with issue must be entered in the textbox field labelled as “BRAND”
7. Warranty status of the product with issue to be entered in the textbox field labelled as “WARRANTY STATUS”
8. Model of the product with issue to be entered in the textbox field labelled as “MODEL”
9. Serial number of the product with issue to be entered in the textbox field labelled as “SERIAL NUMBER”
10. After entering all the details, the CSA must click the button labelled as “LOG ISSUE”
11. This will return another page with a snapshot of customer details including Name, Mobile, Product as entered by the CSA in the previous page.
12. In the page, the CSA must select the issue category from the dropdown provided and also must set the priority status as high or medium or low from the dropdown box provided in the page.
13. The CSA must also describe the issue in detail as described by the customer in the text box field labelled as “Description of Issue”

FR-008 Raise Ticket:

1. Once the issue details are entered by the CSA, the CSA must click the button “CREATE TICKET” as shown in the below picture.

The screenshot shows a web interface for 'ZENITH.SAPPHIRE'. At the top, it says 'Logged-in user : Agent Name'. Below this is the section 'CUSTOMER LOGGING'. It contains a table with customer details, two dropdown menus for 'Issue Category' and 'Priority', a large text area for 'Description of Issue:', and a 'Create Ticket' button with a right arrow.

Field	Value
Name	xxxx
Mobile	xxxxxxxxxx
Product	XXXXXX

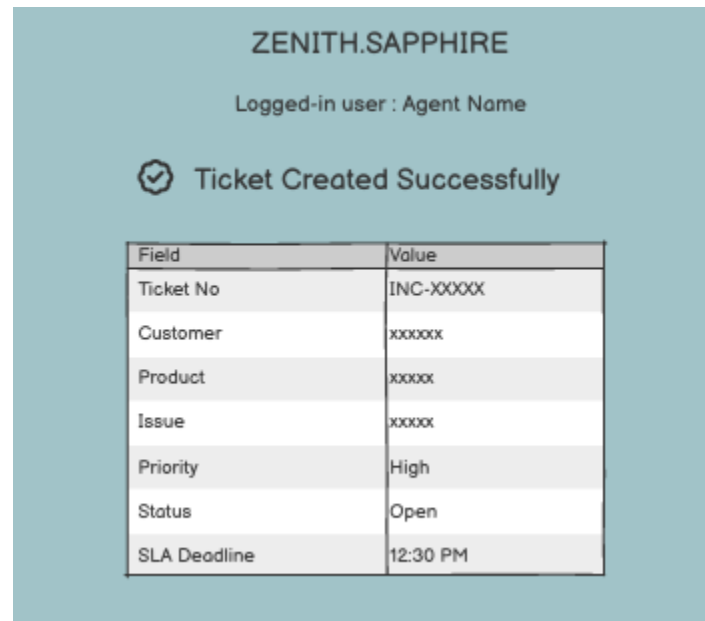
Q Issue Category ▼

Priority ▼

Description of Issue:

Create Ticket >

2. The server must then send a request to the database to generate a ticket number in the format “INC-XXXXX” ,
3. The generated Ticket ID will be displayed to the customer service agent as shown in the picture below and the details associated with the respective Ticket ID must be stored in the database.



FR-009 Troubleshooting:

1. After creation of a ticket, the CSA must be able to access the troubleshooting guide of the respective product. For easy access, the UI must be provided with a dropdown to choose the product category and issue category.
2. Once the product category and issue category are selected and the button "Access troubleshooting content" is clicked, the server must be able to fetch and return the required data from the database.
3. The CSA must guide the customer to solve the issue from the troubleshooting content to the maximum extent possible.
4. If the issue is resolved , the CSA must click the button labelled as "Resolved".
5. The status of the ticket must be changed to Resolved and waiting for approval.
6. The ticket must then move to the team lead for approval.
7. After confirmation, the team lead must approve the resolution by clicking the button "Approve". The ticket status must be changed to "Close" after the team lead's approval.

Logged in user : Agent Name

Available ▾

+9100000000

TICKET INC-XXXX OPEN

PRODUCT CATEGORY ▾

ISSUE CATEGORY ▾

Access Troubleshooting Content

Resolved

Not Resolved

BR-005 Assign Service Centre:

FR-010 Assign to nearest service centre:

1. If the issue is not resolved through troubleshooting, then the CSA must click the button “Not Resolved” must enter the pincode of the customer location in the Text box field “PINCODE”. The pincode is obtained by requesting details from the customer.
2. After entering the pincode, the CSA must click the button labelled as “SEARCH SERVICE CENTRE”
3. The system must already be loaded with master database of all existing customer service centres.
4. The system then search for the service centre nearest to the customer location based on two rules.
 - a. Rule 1: Return the service centre if the entered pin matches exactly with any of the existing service centre’s pincode.
 - b. Rule 2: If the Rule 1 is not satisfied, then return the service centre whose pincode has the shortest distance to the pincode entered by the CSA.
5. Once the CSA gets the display of the nearest service centre, the CSA must select the service centre and click the button labelled as “ASSIGN”.
6. After clicking the button “ASSIGN” the CSA must see a display that “Ticket is successfully assigned to the service centre”
7. The status of the ticket must now be changed to “Assigned to SC”.

TICKET INC-XXXX OPEN

1

XXXXXXXX
Address

2

XXXXXXXX
Address

3

XXXXXXXX
Address

4

XXXXXXXX
Address

5

XXXXXXXX
Address

FR-011 Service Centre Status Update:

1. After a ticket is assigned to the service centre by a CSA, the ticket will be linked to the selected service centre.
2. The service centre must be able to view the assigned tickets from their dashboard under the menu "ASSIGNED TICKETS"
3. On clicking the assigned tickets button, list of all assigned tickets must be displayed on the UI.
4. The service centre must select a particular ticket. On selecting a ticket, two options must be shown as "ACCEPT" and "DECLINE"
5. The service centre must select either "ACCEPT" or "DECLINE"

Logged in user : Service Centre Name

DASHBOARD

ASSIGNED TICKETS

TROUBLE SHOOTING

NOTIFICATIONS

SETTINGS

Ticket ID	Customer Number	Product	Issue	Priority	Assigned Time	Decision
INC-XXXX						Accept Decline
INC-YYYY						Declined
INC-ZZZZ						Accepted

6. If accepted, the ticket status must be changed to "In progress at SC" and the time of acceptance must be recorded for SLA tracking purpose.
7. Once the issue is resolved at the service centre, then the service centre must click the button "Resolve".
8. The ticket status must then be updated to "CLOSE".
9. If declined, then the service centre must enter the reason for decline in the text box field labelled as "Reason"
10. The CSA UI will be notified as "Service centre declined ticket – Reassignment Required".

FR-012 Ticket Reassignment and Escalation:

1. The system must provide an option to the CSA to reassign the service centre in case of decline by service centre.
2. Such tickets declined by service centres must be placed under the tab labelled as “Tickets Returned” in the CSA dashboard.
3. On clicking the tab “Tickets Returned”, the CSA must see the list of all the tickets returned.
4. The CSA must then select the particular ticket. On selecting the ticket, the CSA must be able to view all the details pertaining to the ticket.
5. The CSA must be given an option to assign other service centre. The CSA must enter the pincode of the customer in the text box field “Customer Location” and click the button “SEARCH SERVICE CENTRE”
6. The server must retrieve data from the data base and return at least five and utmost ten service centres from nearest to farthest along with the contact numbers of those service centres.
7. The CSA must then manually call the service centres and verify if the issue can be rectified at that particular service centre. After confirmation, the CSA must reassign the service centre by selecting the respective service centre displayed and by clicking the button “REASSIGN”
8. If the reassignment of ticket is not possible, then the CSA must escalate the ticket to the supervisor. This is done by clicking the button “ESCALATE”.
9. The ticket must then appear under the tab “Escalated Tickets” in team lead’s dashboard.
10. The ticket status must be changed to “Escalated”.
11. The team lead’s dashboard must be notified as “Needs Attention – Ticket INC-XXXXX Escalated”.

FR-013 Track SLA:

1. The system must start the SLA timer as soon as a ticket is raised with a unique ticket ID in the format INC-XXXXX
2. The SLA tracker must be updated whenever the ticket status changes.
3. The SLA tracker must also display alert message whenever a ticket nears SLA limits.
4. As soon as the customer enters the issue category and raises the ticket, the server must fetch data from database (SLA configuration table) and attach it to the ticket for SLA tracking.
5. The server will display the SLA timer for each tickets of respective CSA’s dashboard.
6. The server must display SLA performance of entire team on Team lead’s dashboard.
7. The server must also display SLA status on the service centre’s dashboard.

BR-006 Ticket Closure:**FR-013 Close Ticket:**

1. Ticket resolution happens in following ways

- a. When the CSA resolves an issue through troubleshooting and click “Resolve”, it must move to the team leader dashboard under Resolved tickets. The team lead must click “Approve” button after verification. The ticket status must then be changed to “CLOSE”. The closed tickets must move to the tab “Tickets Closed” in both CSA and Team lead’s dashboard.
- b. When the Service Centre resolves an issue and click “Resolve”. The ticket status must be changes to “CLOSE”. The closed tickets must move to the tab “Closed Tickets”

Validation Check:

Field Name	Field Type	Data Type	Required Field?	Min. Char.	Max Char.	Notes
Name	Text Box	Alpha	Yes	2	30	“Entered Name is Invalid”
Mobile No	Text Box	Numeric	Yes	10	10	This field to capture customer no for validation & registration.
Password	Text Box	A) 1 Capital letter B) 1 Small Letter C) 1 Numeric D) 1 special character	Yes	8	15	Data is encrypted
Re-enter Password	Text Box	A) 1 Capital letter B) 1 Small Letter C) 1 Numeric D) 1 special character	Yes	8	15	Data is encrypted